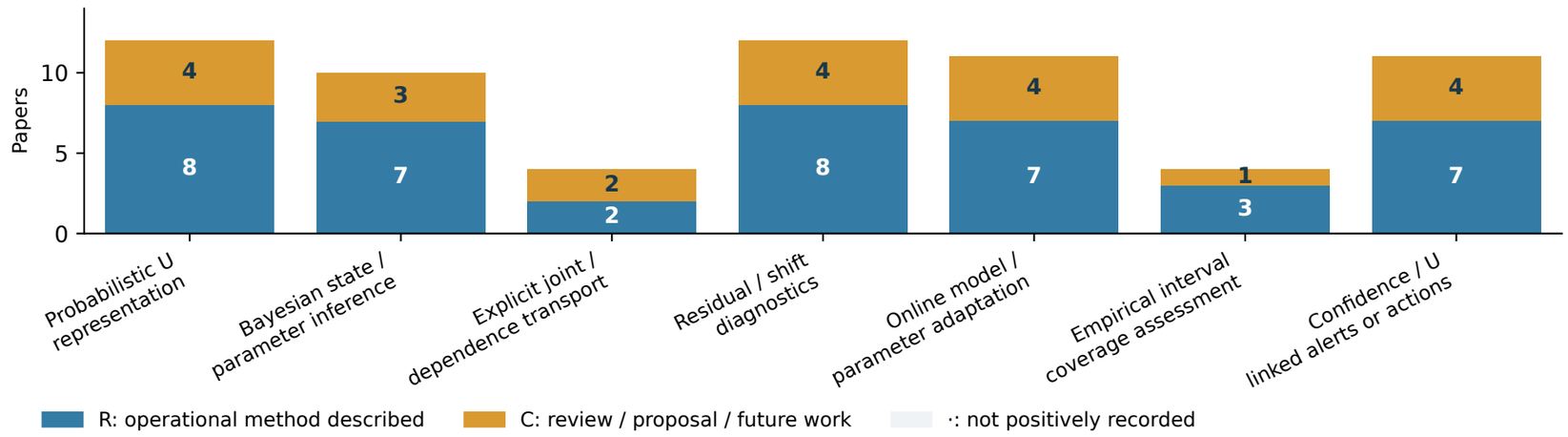
03 Digital Twin architectures × uncertainty mechanisms

Thirteen archived appraisals. Method description, conceptual support and missing positive coding remain separate.

		Probabilistic U representation	Bayesian state / parameter inference	Explicit joint / dependence transport	Residual / shift diagnostics	Online model / parameter adaptation	Empirical interval coverage assessment	Confidence / U linked alerts or actions
Review / reference architecture	001 Physics-informed review	C	C	C	C	C	C	C
	009 Soft-sensor review	C	·	·	C	C	·	C
	010 Water-quality review	C	C	C	C	·	·	C
Hybrid model fusion	002 Hybrid fuel-system twin	R	R	·	R	R	·	R
Inference component	004 Conditional state–parameter inference	R	R	R	·	R	·	·
	008 Self-protection architecture	·	·	·	R	R	·	·
Feedback / control	015 Conflict-aware O-RAN control	R	·	·	R	R	·	R
	016 Parallel twins for NMPC	R	R	R	R	R	·	R
Layered monitoring	011 Bio360 event framework	C	C	·	R	·	·	R
	012 Field water-quality twin	R	R	·	R	C	R	R
	018 CNC tool-health twin	R	R	·	R	R	R	R
Bayesian learner	013 Hierarchical posterior evolution	R	R	·	R	R	R	R
	014 Adaptive physical-prior BNN	R	R	·	C	C	·	C



R does not mean independently validated. 004 is a preprint inference component; 015 is an adjacent control system.

Joint/dependence transport is coded for the specified variables only. It does not establish end-to-end DT continuity.